



ESP32-S3 Project

Gesture-controlled smart camera

DIY Gesture-Controlled Smart Camera

Hardware Setup & Connectivity

The Core Components



Shared SPI Bus Wiring
The TFT and SD module share the D8 (SCK) and D10 (MOSI) pins.



Component	Pin: SCK (SCL)	Pin: MOSI (SDA)	Pin: CS
TFT Display	XIAO D8	XIAO D10	XIAO D1
SD Card	XIAO D8	XIAO D10	XIAO D4

Gesture Interface Logic



Capture Photo (📷)
Triggered when a large hand is detected close to the camera lens.



Start/Stop Video (📹)
Movement detection or a hand wave toggles video recording on and off.



Preview Mode
The system defaults to a live preview when no hand is detected.

Team members:

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Internal Guide:

Mr. R. Priyakanth



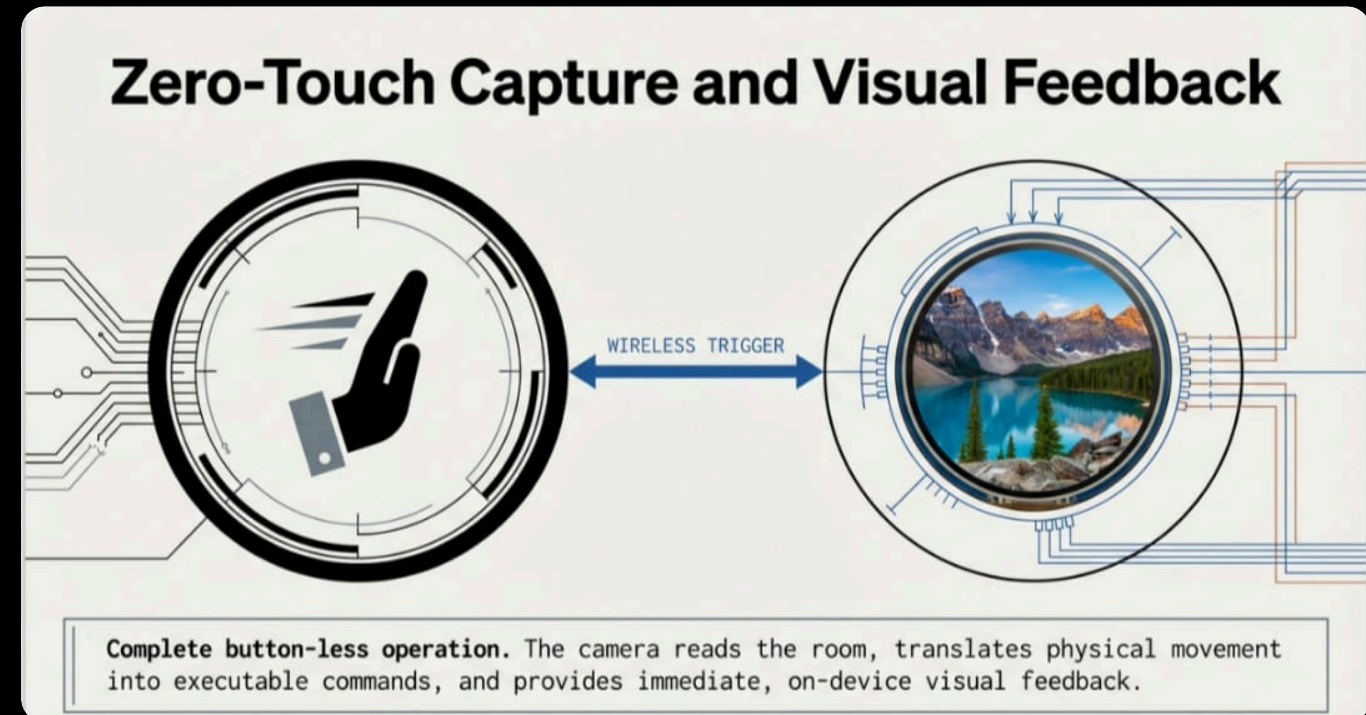
Project aim

Design and implement a smart camera system that uses real-time gesture recognition to control image capture and video recording - entirely without physical buttons.

A self-contained embedded prototype built on the XIAO ESP32-S3 Sense, demonstrating a practical, hands-free IoT camera.

Key goals

- Detect hand gestures via the onboard camera
- Show live preview on a GC9A01 240×240 round TFT display
- Store photos and video on a 32GB MicroSD card

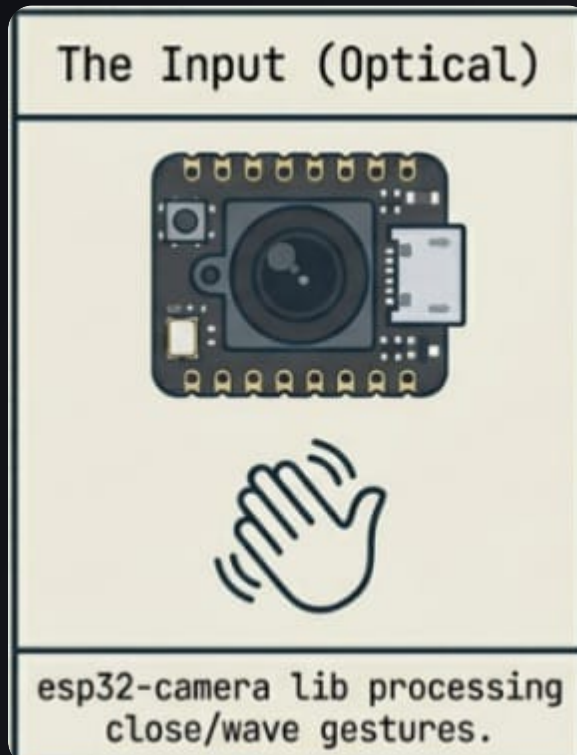




Hardware components

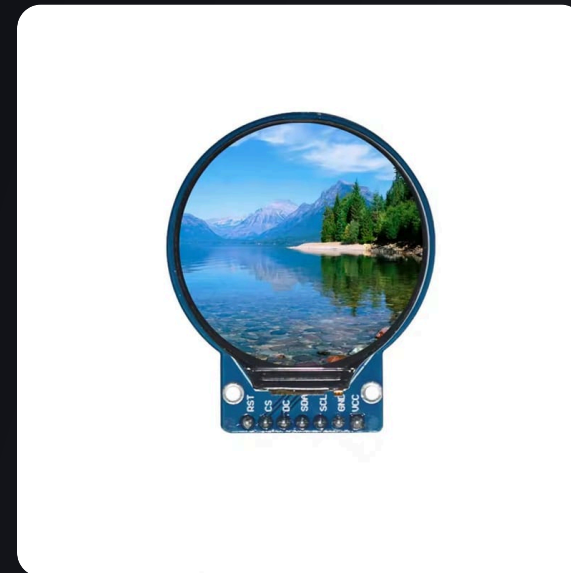
XIAO ESP32S3 Sense

Dual-core MCU at 240MHz.
Onboard OV2640 camera
and mic, Wi-Fi, Bluetooth,
8MB PSRAM and 8MB Flash.



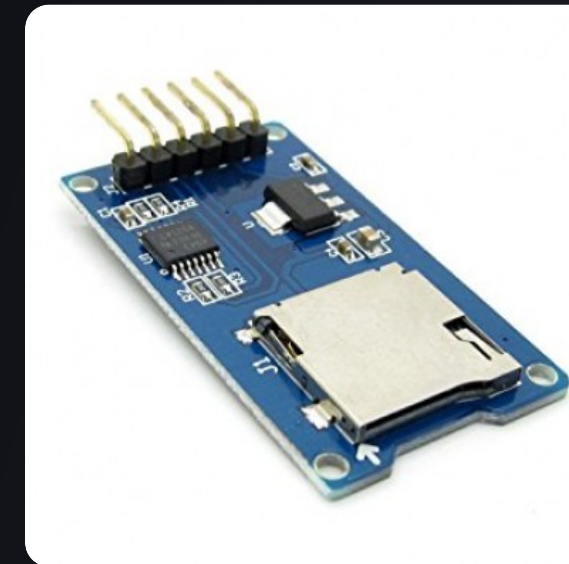
GC9A01 Round Display

240×240 circular SPI display
for live camera preview.
Connected via MOSI, SCLK,
CS, DC and RST pins.



MicroSD Module

SPI-based storage module
for writing JPEG photos and
video files captured by the
camera.



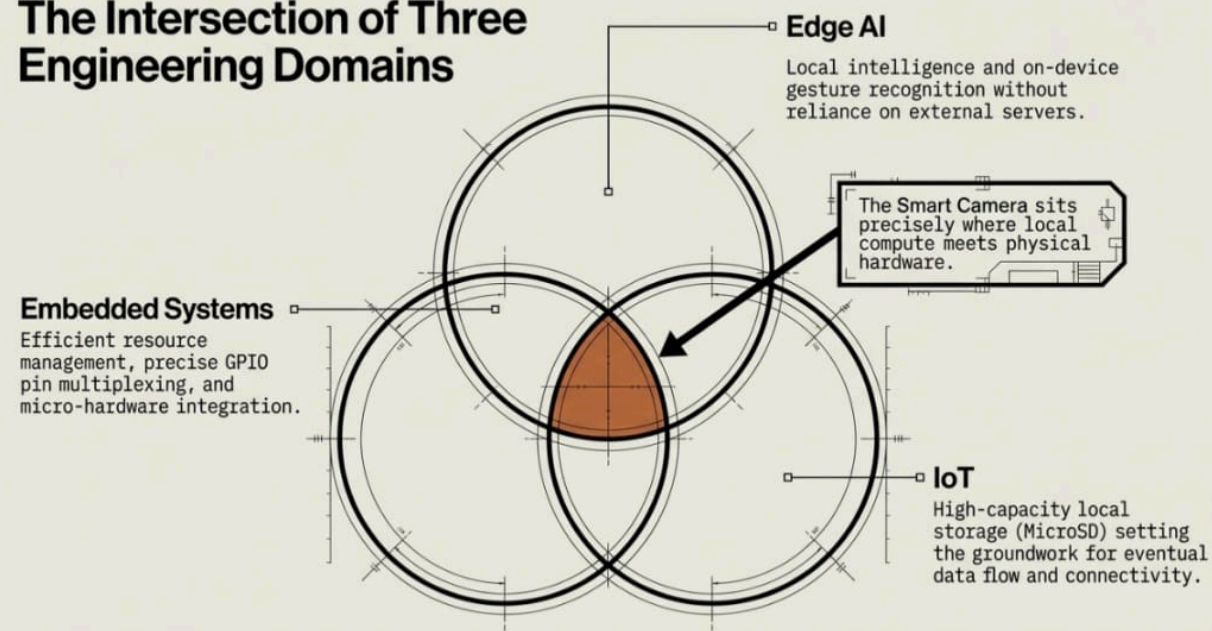
32GB MicroSD Card

Stores photos and video
clips. FAT32-formatted for
full compatibility with Arduino
SD libraries.



Four domains, one system

The Intersection of Three Engineering Domains



This project bridges hardware, vision, inference, and interaction into a single embedded platform.

Embedded Systems

ESP32-S3 dual-core 32-bit MCU with built-in Wi-Fi and Bluetooth - the processing heart of the device.

Computer Vision

Camera-based gesture recognition using pixel-level motion and proximity analysis - no external sensor needed.

IoT and Edge AI

On-device inference with no cloud dependency - all processing runs locally on the ESP32-S3.

Human-Computer Interaction

Natural gestures replace physical buttons for a more intuitive user experience.



Dev environment and libraries



Dev environment and libraries



Arduino IDE

Primary dev environment for firmware writing, and flashing firmware.

ESP32 Arduino Core



Board support package for coding ESP32-Bo hardware within



Camera and display

Camera driver and TFT display rendering



Storage and FS

SD card access and file system management

Arduino IDE

Primary dev environment for writing, compiling, and flashing firmware to the XIAO ESP32S3 Sense.

ESP32 Arduino Core

Board support package enabling ESP32-S3 peripherals - camera, SPI, and SD card - within the Arduino ecosystem.

Camera and display

esp_camera.h - OV2640 camera driver
TFT_eSPI / Arduino_GFX - TFT display rendering

Storage and FS

SD.h / SD_MMC.h - MicroSD read/write
FS.h - File system abstraction layer



Design Methodology

Design Methodology

01 Hardware Setup



- Wire GC9A01 TFT and MicroSD to ESP32-S3 via SPI. Power the XIAO board via USB-C.

02 Firmware Development

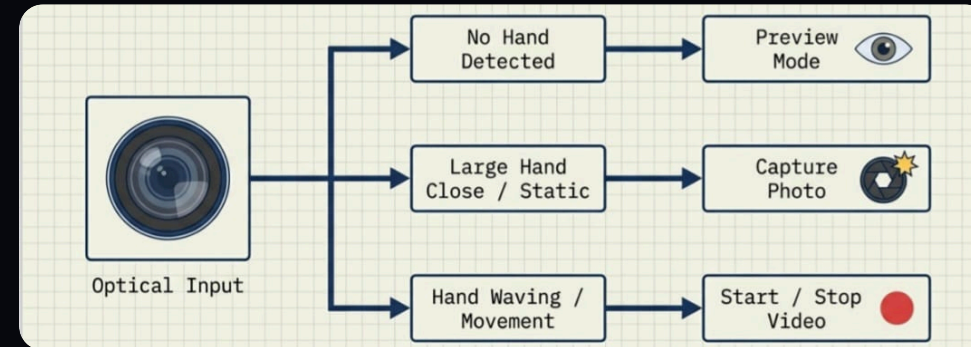


- Initialize OV2640, TFT, and SD card. Build gesture detection, JPEG capture, and frame write functions.

03 Testing & Calibration



- Calibrate gesture thresholds under varied lighting.
- Verify SD write speed and validate file storage structure.



Hardware Architecture (Shared SPI Bus)

The system uses a shared SPI bus where the TFT and SD modules share SCK (D8) and MOSI (D10) pins. Independent Chip Select (CS) pins—D1 for the display and D4 for storage—manage data traffic and conserve GPIOs.

Interface Logic (Zero-Contact Gestures)

Buttons are replaced by Edge AI gesture recognition:

a 🖐️ large hand triggers photos and

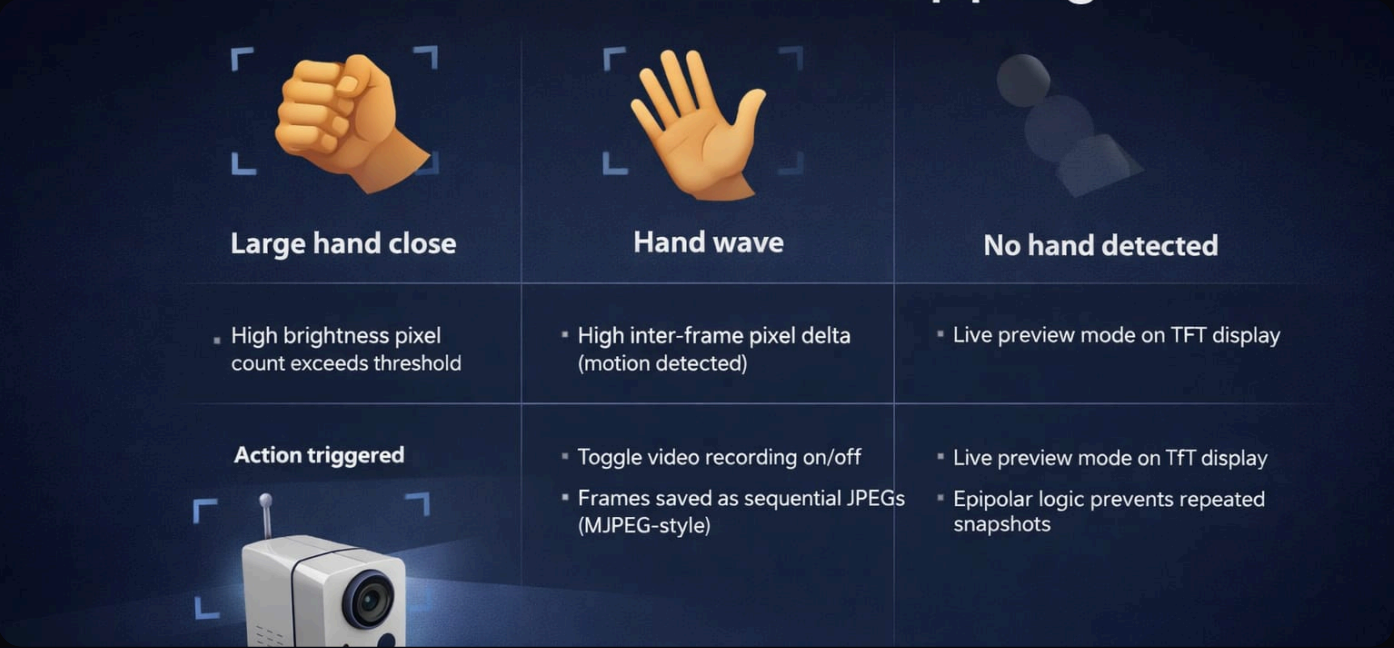
a 🖐️ wave toggles video. The system defaults to a live preview mode on the circular TFT when no hand is detected.

Software & Offline Processing Developed in Arduino IDE using esp32-camera and TFTeSPI libraries for high-speed local processing. The design prioritizes offline Edge AI, performing all vision and storage tasks directly on the ESP32-S3 without Wi-Fi.



Gesture-to-action mapping

	👊 Large hand close	👋 Hand wave	No hand detected
Detection method	1.High brightness pixel count exceeds threshold	1.High inter-frame pixel delta (motion detected)	1.Low brightness and low motion across frame
Action triggered	1.Capture JPEG photo - save to SD card 2.2-second cooldown after each capture	1.Toggle video recording on/off 2.Frames saved as sequential JPEGs (MJPEG-style)	1.Live preview mode on TFT display 2.Debounce logic prevents repeated triggers





Future scope

What comes next

Wi-Fi cloud upload

Transmit captured images to a cloud endpoint or local server via the ESP32-S3's builtin Wi-Fi.

Pairs with LiPo battery for fully wireless, portable operation.

TensorFlow Lite ML

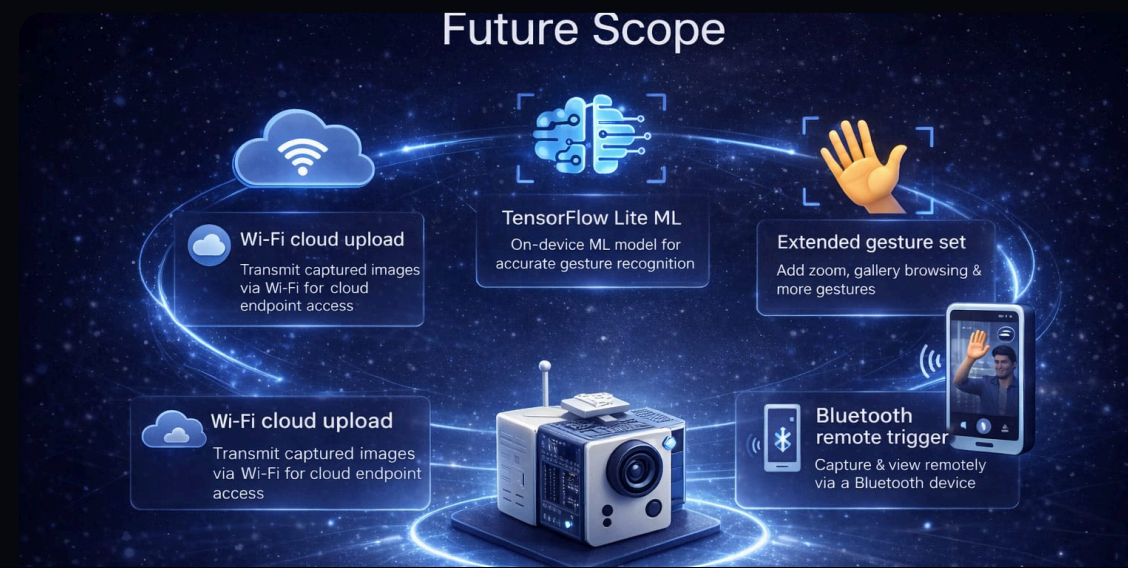
Replace threshold-based detection with a lightweight on-device ML model for more accurate, diverse gesture recognition without a network dependency.

Extended gesture set

Add gestures for zoom, brightness, filter selection, and gallery browsing. Display capture count and recording timer on the TFT screen via OLED/TFT UI overlay.

Bluetooth remote trigger

Allow a paired Bluetooth device to trigger capture or view a live stream wirelessly - removing the need for physical proximity to the camera.





References



■ Technical References

- Espressif Systems - ESP32-S3 Datasheet & Documentation
- Arduino IDE Official Documentation
- ESP32 Arduino Core GitHub Repository
- OpenCV Documentation (for gesture concepts)
- TensorFlow Lite (for future ML integration concepts)

■ Research & Concept References

- Gesture Recognition Techniques in Computer Vision (IEEE Papers)
- Embedded AI and Edge Computing Concepts
- Human-Computer Interaction using Vision-Based Systems

■ Online Resources

- Espressif Official Website: <https://www.espressif.com>
- Arduino Official Website: <https://www.arduino.cc>

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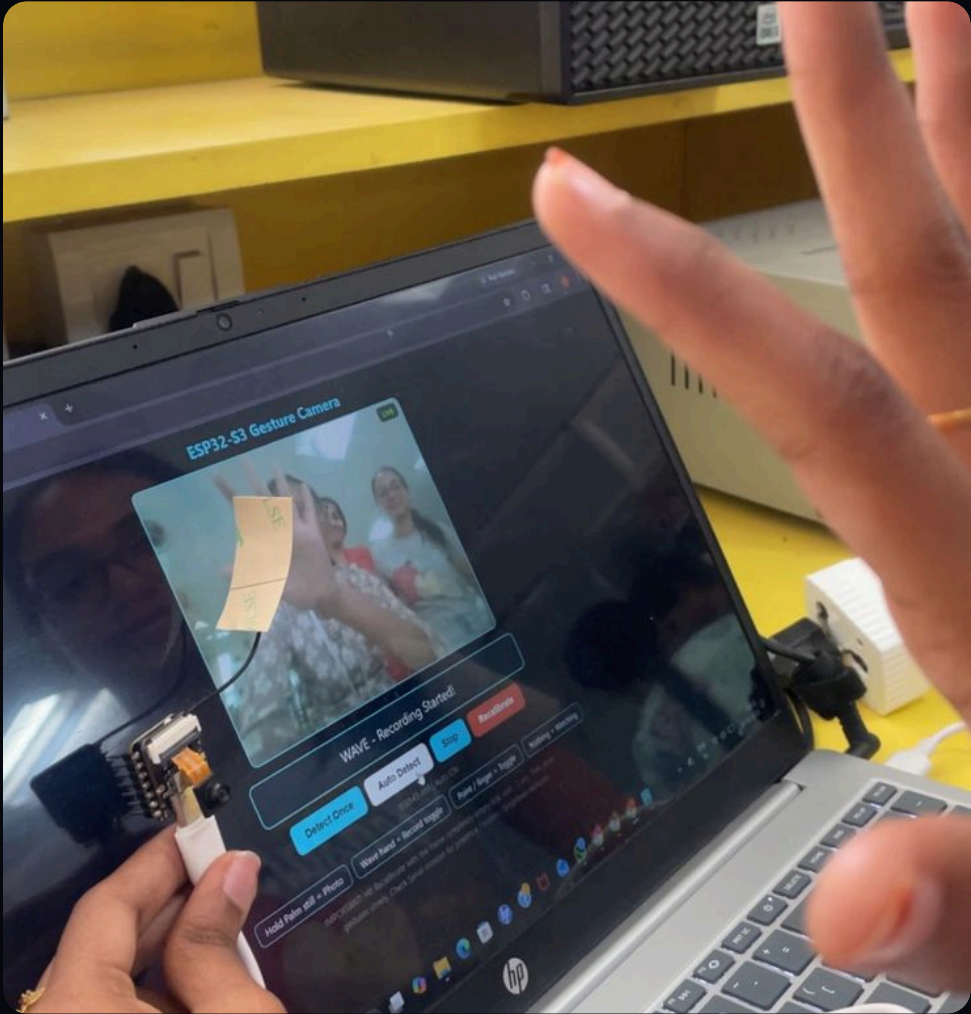
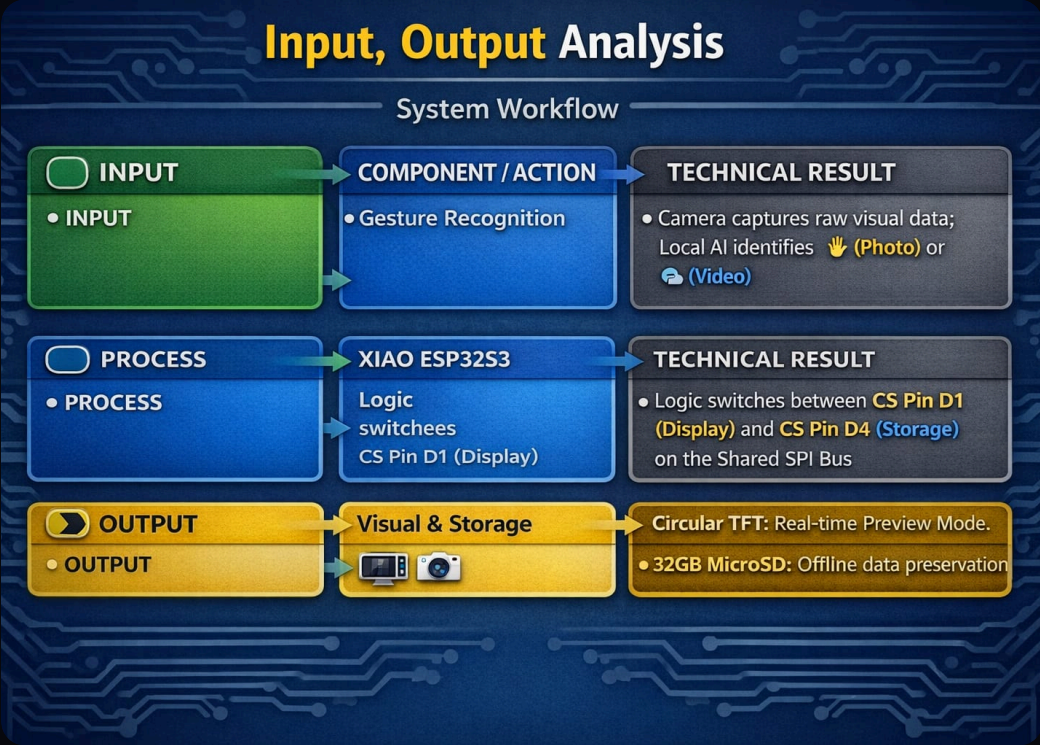
■ Online Resources

- Espressif Official Website: <https://www.espressif.com>
- Arduino Official Website: <https://www.arduino.cc>
- Random Nerd Tutorials (ESP32 Guides)
- GitHub Repositories for ESP32 Camera Projects

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RESULT



Thank you

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